

Ali Abouelazm

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Education

Texas A&M University

Aug. 2023 – May 2028

B.S. Data Engineering | Minor in Mathematics

College Station, TX

- Relevant Coursework: Data Structures & Algorithms, Quant Models for Machine Learning, Statistical Learning & Decisions, Optimization of Analytics, Computational Data Science, Principles of Data Science, Statistics.

Experience

Tesla

Aug. 2026 – Dec. 2026

Data Engineer Intern, Fleet Analytics

Palo Alto, CA

- Incoming Fall 2026, building data pipelines and analytics tooling on **petabyte-scale** vehicle telemetry from millions of Tesla vehicles, enabling fleet-wide performance monitoring and data-driven insights across the Fleet Analytics team.

Cloudflare

May 2026 – Aug. 2026

Data Science Intern

Austin, TX

- Pioneered a **Narrative Mismatch Engine**, a novel approach in the **AEO** space, ingesting **3 years** of **SEMrush** data across **50K+** AI queries and thousands of pages to detect gaps between AI perception and intended product positioning.
- Applied **NLP** and **embedding-based** semantic analysis to surface narrative drift across **CDN**, **Zero Trust**, and **SASE** product lines, enabling content teams to implement targeted changes that close AI perception gaps at scale.
- Increased **AI content visibility** and narrative alignment across product marketing surfaces by driving data-informed updates from query-level mismatch signals extracted from large-scale web and AI traffic data.

Texas A&M AgriLife

Mar. 2026 – Present

Machine Learning Researcher

College Station, TX

- Building **PyTorch**-based predictive models on **50GB+** of livestock biosensor data to forecast animal health indicators across **3+** active deployments, iterating continuously on model architecture, feature design, and training pipelines.
- Architecting scalable **ML** pipelines on **AWS** (S3, Athena) to ingest and process raw biological data streams, cutting preprocessing time by **35%** and reducing time-to-insight for field researchers and enabling faster iteration.
- Implementing automated feature engineering to extract health signals from noisy multi-sensor inputs, improving model training throughput by removing manual preprocessing steps and increasing prediction reliability across all deployments.

TCG Digital Solutions

May 2025 – Aug. 2025

Data Science Intern

Somerset, NJ

- Architected an end-to-end **CV** pipeline using **Gemini Pro** and **OpenCV** to automate soccer highlight extraction, hitting **95%** precision across **100+** match hours and reducing manual processing by **80%**; tuned **XGBoost** and **CatBoost** models on **1M+** rows of player tracking data using **SHAP** values to surface tactical insights.

Projects

Interview Mirror | PyTorch, FastAPI, React, Whisper, MediaPipe, WebSockets | mirror.aliabouelazm.com

May 2026

- Trained two emotion **CNNs** from scratch on **MPS**: a face model on **35K FER+** images (**0.78** accuracy, **0.67** F1) with MixUp, CutMix, and EMA; a voice model on **9.4K** audio clips (**0.67** F1) with SpecAugment and mel-level MixUp.
- Built a **21-feature** fusion **MLP** combining face softmax, voice softmax, acoustic descriptors, and language signals (filler/hedge density) into continuous confidence and engagement scores, running end-to-end at **500ms** via **WebSocket**.

Sonus | Python, FastAPI, React, scikit-learn, SQLite, WebSockets, OpenAI API

Feb. 2026

- Built an autonomous **LLM** assistant that reasons over calendar events, biometrics, and live device state to execute complex multi-step routines unprompted, orchestrating **10+** service integrations via a tool-calling loop.
- Trained personalized stress and sleep classifiers on biometric data from **3** wearable integrations using **scikit-learn**, with a confidence scoring system that learns from user feedback and adapts via real-time **WebSocket** loops.

Drift | Python, FastAPI, React, DistilBERT, HuggingFace, SHAP | drift.aliabouelazm.com

Apr. 2026

- Fine-tuned **DistilBERT** on **40K+** labeled Reddit and Twitter posts for 3-class sentiment classification, achieving **73.1%** F1 macro on a **10,360**-sample held-out test set across positive, negative, and neutral sentiment classes.
- Built an analytics engine computing rolling sentiment averages across configurable time windows, anomaly detection with mild/moderate/severe severity scoring, **TF-IDF** keyword extraction per class, and **SHAP** token-level explanations.

Leadership

Lovable.dev | TAMU Datathon

Jan. 2026 – Present

Campus Ambassador | Challenges Organizer

Texas A&M University

- Hosted campus-wide AI platform workshops for **200+** engineering students as Lovable Campus Ambassador; directed **ML** problem set creation for **500+** competing students at TAMU Datathon using **Python**, **SQL**, and **scikit-learn**.

Technical Skills

ML & AI: PyTorch, TensorFlow, scikit-learn, XGBoost, DistilBERT, SHAP, Whisper, MediaPipe, LLM Tool Calling (OpenAI/Gemini APIs), OpenCV.

Languages: Python, SQL, C/C++, TypeScript, JavaScript, R.

Data & Cloud: AWS (S3, Athena), PostgreSQL, Snowflake, Docker, Git, FastAPI, WebSockets.

Libraries: pandas, NumPy, HuggingFace Transformers, Matplotlib, Seaborn, Dask, SciPy.